NETWORK INFRASTRUCTURE PRACTICAL MANUAL

Module: Network Infrastructure
Programme: Information Technology / Networking Support
Practical Assessments: 9
Total Marks: 100
Lab Size: 10 Student Workstations

1. MODULE OVERVIEW

This practical module equips students with the knowledge and skills required to design, install, configure, manage, test, and troubleshoot network infrastructure in a modern enterprise environment.

The practical exercises progress from basic cabling and hardware installation to Cisco router and switch configuration, VLAN implementation, network services, and troubleshooting.

2. AVAILABLE EQUIPMENT

Structured Cabling Equipment

Item	Quantity
Crimping Tools	4
Punchdown Tools	4
Patch Panels	4
Network Patch Boxes	13
0.5m Fly Leads	13
1m Fly Leads	13
Network Cable (500m)	4 Rolls

Cisco Equipment

Item	Quantity
Cisco Console Cables	4
Cisco 1941 Routers	2
Cisco ISR 4331 Routers	2
Cisco Catalyst 3560 Switches	2

Network Infrastructure Equipment

Item	Quantity

24-Port Switches	4
24-Port Patch Panels	4
4U Switch Cabinets	4

Computer Equipment

Item	Quantity
Lenovo Desktop PCs	10
LG Screens	10
Keyboard & Mouse Sets	10

Printing Equipment

Item	Quantity
HP LaserJet 600 M602 Printer	1
HP Color LaserJet 5500DN Duplex Printer	1
HP Colour LaserJet Professional CP5225N Printer	1

Toolkits

Item	Quantity
SPROTEK 58-Piece Computer Technician Toolkit	2
SPROTEK 10-Piece Network Installation Toolkit	2

Classroom Equipment

Item	Quantity
Projector	1
Projector Mounting	1
Whiteboard	1

3. LEARNING OUTCOMES

Upon successful completion of this module, students will be able to:

1. Safely use networking tools and equipment.
2. Construct Ethernet cables using industry standards.
3. Install and terminate patch boxes and patch panels.
4. Build structured cabling systems.
5. Install and manage switch cabinets.
6. Access and configure Cisco IOS devices.

7. Configure Cisco switches and routers.
8. Implement VLANs and network segmentation.
9. Configure shared network resources.
10. Troubleshoot enterprise network faults.
11. Document network installations professionally.

PRACTICAL 1: NETWORK TOOLS AND SAFETY

Objective

Identify networking tools and apply safe working procedures.

Activities

1. Identify networking tools.
2. Explain the use of each tool.
3. Demonstrate safe handling.
4. Complete a tool inventory sheet.
5. Apply health and safety procedures.

Deliverable

Completed Tool Identification Worksheet.

Assessment Rubric (10 Marks)

Criteria	Marks
Tool identification	4
Understanding tool functions	2
Safe tool usage	2
Worksheet completion	2
Total	**10**

PRACTICAL 2: ETHERNET CABLE CONSTRUCTION

Objective

Construct and test Ethernet patch cables using T568B.

Activities

1. Measure and cut cable.
2. Strip cable jacket.

3. Arrange conductors according to T568B.
4. Crimp RJ45 connectors.
5. Test completed cables.

T568B Wiring Standard

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Deliverable

Three tested Ethernet cables.

Assessment Rubric (10 Marks)

Criteria	Marks
Correct cable sequence	4
Cable preparation	2
Crimp quality	2
Connectivity testing	2
Total	**10**

PRACTICAL 3: PATCH BOX INSTALLATION

Objective

Install and terminate network outlets.

Activities

1. Install patch boxes.
2. Terminate using T568B.
3. Label outlets.
4. Test connectivity.

Deliverable

Fully functional network outlet.

Assessment Rubric (10 Marks)

Criteria	Marks
Correct termination	4
Use of punchdown tool	2
Cable management	2
Labelling	2
Total	**10**

PRACTICAL 4: PATCH PANEL INSTALLATION

Objective

Install and document a patch panel.

Activities

1. Mount patch panel.
2. Terminate network runs.
3. Label ports.
4. Create port schedule.
5. Test all ports.

Deliverable

Completed patch panel installation.

Assessment Rubric (10 Marks)

Criteria	Marks
Correct patch panel termination	4
Port numbering	2
Documentation	2
Testing verification	2
Total	**10**

PRACTICAL 5: CISCO DEVICE ACCESS AND BASIC CONFIGURATION

Objective

Access Cisco IOS and perform basic device configuration.

Equipment

- Console Cables
- Cisco 1941 Router
- Cisco ISR4331 Router
- Cisco Catalyst 3560 Switch

Activities

1. Access Cisco devices using console cables.
2. Configure hostname.
3. Configure passwords.
4. Configure login banner.
5. Save configurations.

Example Configuration

```
enable
configure terminal

hostname POTCH-R1

enable secret Cisco123

line console 0
password Cisco123
 login

 banner motd # Unauthorized Access Prohibited #

 end
 copy running-config startup-config
```

Deliverable

Configuration screenshots and configuration file.

Assessment Rubric (10 Marks)

Criteria	Marks
Console connectivity	2
IOS navigation	3
Device configuration	3
Save and verify configuration	2
Total	**10**

PRACTICAL 6: LAN CONSTRUCTION AND SWITCH CONFIGURATION

Objective

Build a LAN using a Cisco Catalyst switch.

Equipment

- 10 PCs
- Cisco Catalyst 3560 Switch
- Patch Panel
- Fly Leads

Activities

1. Connect all 10 PCs.
2. Configure static IP addresses.
3. Configure switch management interface.
4. Verify connectivity.
5. Create a network diagram.

IP Addressing Scheme

Device	IP Address
PC1	192.168.10.1
PC2	192.168.10.2
PC3	192.168.10.3
PC4	192.168.10.4
PC5	192.168.10.5
PC6	192.168.10.6
PC7	192.168.10.7
PC8	192.168.10.8
PC9	192.168.10.9
PC10	192.168.10.10
Switch VLAN 1	192.168.10.254

Subnet Mask: **255.255.255.0**

Verification Commands

```
show running-config
show vlan brief
show mac address-table
```

Deliverable

Operational LAN and topology diagram.

Assessment Rubric (15 Marks)

Criteria	Marks
Physical installation	3
Switch configuration	4
IP configuration	4
Connectivity testing	4
Total	**15**

PRACTICAL 7: ROUTER CONFIGURATION AND INTER-NETWORK COMMUNICATION

Objective

Configure routers and establish communication between networks.

Equipment

- Cisco 1941 Router
- Cisco ISR4331 Router
- Cisco Switches
- PCs

Activities

Create two networks:

Network A

```
192.168.10.0/24
```

Network B

```
192.168.20.0/24
```

Configure router interfaces.

Example Configuration

```
interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown

```

```
interface g0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
```

Verification Commands

```
show ip interface brief
ping
tracert
```

Deliverable

Successful communication between both networks.

Assessment Rubric (10 Marks)

Criteria	Marks
Router configuration	4
Interface addressing	2
Connectivity testing	2
Documentation	2
Total	**10**

PRACTICAL 8: VLAN IMPLEMENTATION AND NETWORK SEGMENTATION

Objective

Create VLANs and segment network traffic.

Equipment

- Cisco Catalyst 3560 Switches
- 10 PCs

VLAN Design

VLAN	Department	Devices
VLAN 10	Administration	PC1-PC3
VLAN 20	Lecturers	PC4-PC6
VLAN 30	Students	PC7-PC10

Activities

1. Create VLANs.
2. Assign ports to VLANs.

3. Verify VLAN membership.
4. Test communication.
5. Document VLAN design.

Example Commands

```
vlan 10
name ADMIN

vlan 20
name LECTURERS

vlan 30
name STUDENTS
```

Verification

```
show vlan brief
```

Deliverable

Working segmented network.

Assessment Rubric (15 Marks)

Criteria	Marks
VLAN creation	4
Port assignment	4
Connectivity verification	4
Documentation	3
Total	**15**

PRACTICAL 9: ENTERPRISE NETWORK DEPLOYMENT AND TROUBLESHOOTING

Objective

Deploy and troubleshoot a complete Cisco-based enterprise network.

Equipment

- Cisco Routers
- Cisco Switches
- Switch Cabinets
- Patch Panels

- Printers
- 10 PCs

Activities

1. Install switch cabinet.
2. Install patch panels.
3. Configure switches.
4. Configure routers.
5. Implement VLANs.
6. Configure printers.
7. Test and troubleshoot the network.

Introduced Faults

- Incorrect VLAN assignment
- Incorrect subnet mask
- Incorrect default gateway
- Duplicate IP address
- Faulty patch lead
- Disabled switch port
- Router interface shutdown
- Missing route

Troubleshooting Commands

```
show running-config
show vlan brief
show interfaces
show ip route
show ip interface brief
show mac address-table
ping
tracert
```

Deliverable

Completed troubleshooting report.

Assessment Rubric (10 Marks)

Criteria	Marks
Fault identification	3
Troubleshooting methodology	3
Correct resolution	2
Professional report	2

Total	10

PRACTICAL OBSERVATION CHECKLIST

Competency	Yes	No
Uses tools safely	☐	☐
Uses crimping tools correctly	☐	☐
Uses punchdown tools correctly	☐	☐
Applies T568B correctly	☐	☐
Configures Cisco devices correctly	☐	☐
Demonstrates troubleshooting skills	☐	☐
Maintains clean workstation	☐	☐
Demonstrates teamwork	☐	☐
Produces accurate documentation	☐	☐

FINAL CAPSTONE PROJECT

Scenario

Potchefstroom Campus requires a fully operational networking laboratory for 10 workstations. Student teams must design, install, configure, test, secure, document, and troubleshoot the complete network infrastructure.

Infrastructure Requirements

End User Devices

- 10 Desktop PCs
- 10 Monitors
- 10 Keyboard and Mouse Sets

Cisco Equipment

- 2 Cisco Catalyst 3560 Switches
- 2 Cisco 1941 Routers
- 2 Cisco ISR 4331 Routers
- 4 Cisco Console Cables

Infrastructure Equipment

- 4 Patch Panels

- 4 Switch Cabinets
- 13 Network Outlets
- Structured Cabling System
- Fly Leads

Shared Resources

- 3 Network Printers
- Projector
- Whiteboard

Project Tasks

Phase 1: Planning

- Produce physical topology
- Produce logical topology
- Create IP addressing plan
- Design VLAN structure

Phase 2: Infrastructure Installation

- Install switch cabinets
- Install patch panels
- Terminate outlets
- Test cabling

Phase 3: Cisco Switch Configuration

- Configure hostnames
- Configure passwords
- Configure management interfaces
- Configure VLANs

Phase 4: Cisco Router Configuration

- Configure interfaces
- Configure routing
- Verify routing tables
- Test communication

Phase 5: Shared Resource Configuration

- Configure network printers
- Verify access from all VLANs

Phase 6: Testing and Troubleshooting

- Verify connectivity
- Verify VLAN functionality
- Verify routing
- Resolve instructor-introduced faults

Phase 7: Documentation

Submit:

- Network topology diagram
- IP addressing table
- VLAN allocation table
- Switch configurations
- Router configurations
- Connectivity test results
- Troubleshooting report

CAPSTONE PROJECT RUBRIC (20 MARKS)

Assessment Area	Marks
Network Design and Planning	4
Structured Cabling Installation	3
Patch Panel and Cabinet Configuration	2
Cisco Switch Configuration	3
Cisco Router Configuration	3
VLAN Configuration	2
Testing and Troubleshooting	2
Documentation and Professionalism	1
Total	**20**

ASSESSMENT SUMMARY

Assessment	Marks
Practical 1	10
Practical 2	10
Practical 3	10
Practical 4	10
Practical 5	10
Practical 6	15
Practical 7	10

Practical 8	15
Practical 9	10
Total Continuous Assessment	**100**

ACHIEVEMENT LEVELS

Percentage	Achievement
80 - 100%	Excellent
70 - 79%	Competent
60 - 69%	Satisfactory
50 - 59%	Basic Competence
Below 50%	Not Yet Competent

Expected Graduate Competencies

Students who successfully complete this practical programme will be able to:

- Install structured cabling systems.
- Configure patch panels and switch cabinets.
- Configure Cisco Catalyst switches.
- Configure Cisco 1941 and ISR4331 routers.
- Implement VLANs and network segmentation.
- Configure network printers and shared resources.
- Troubleshoot physical and logical network issues.
- Deploy and document a small enterprise network consisting of 10 workstations using industry-standard networking practices.